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The sin-omial integral for $\binom{r}{s}$: a one-page elementary proof

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Theorem 1. *For real $r > 0$ and $0 \le s \le r$,*

$$\frac{1}{\pi}\int_0^\pi \frac{\sin^r x}{\sin^s\left(\frac{s}{r}x\right)\,\sin^{r-s}\left(\frac{r-s}{r}x\right)}\,\mathrm{d}x = \binom{r}{s} = \frac{\Gamma(r+1)}{\Gamma(s+1)\,\Gamma(r-s+1)}.$$

Setup. For $s \in \{0, r\}$ both sides are 1 ($\sin^0 := 1$; $r = 0$ is excluded, as s/r is undefined). Let $0 < s < r$, $a = s/r$, $b = 1-a$, and on $[0,\pi]$ let $R = \frac{\sin\theta}{\sin b\theta}$, $\rho = \frac{\sin a\theta}{\sin b\theta}$, $f = R\rho^{-a}$, with the limits $R(0) = \frac{1}{b}$, $\rho(0) = \frac{a}{b}$. As $\sin(c\theta)/\theta$ is entire, and positive on $[0,\pi]$ if $0 < c < 1$, $R \ge 0$ and $\rho > 0$ are smooth and f is continuous on $[0,\pi]$; $\rho(\pi) = 1$ as $\sin b\pi = \sin a\pi$. As $f = \sin\theta/(\sin^a a\theta\,\sin^b b\theta)$ on $(0,\pi)$, the integrand is $f(x)^r$. Write $\operatorname{sinc} u = \frac{\sin \pi u}{\pi u}$, $\operatorname{sinc} 0 = 1$, and $\binom{n}{\kappa} = \frac{\Gamma(n+1)}{\Gamma(\kappa+1)\Gamma(n-\kappa+1)}$ ($1/\Gamma$ is entire).

Idea. On the loop $z = \rho(\theta)e^{i\theta}$ the triangle $(-1, 0, z)$ has angles $a\theta, \pi-\theta, b\theta$, so $(1+z)^n z^{-an} = f(\theta)^n$ is real, with $z^{-an} := \rho^{-an}e^{-ian\theta}$. So f^n is the integrand of Cauchy's integral $\frac{1}{2\pi}\int_{-\pi}^{\pi}(1+e^{it})^n e^{-iant}\,\mathrm{d}t = \binom{n}{an}$, moved from the unit circle onto this loop. For integer n it is $\sum_j \binom{n}{j} z^{j-an}$, with explicit primitives. Real exponents: by $(*)$ below, the integer case says that $f(X)\,T^a(1-T)^b$ ($X \sim U(0,\pi)$, $T \sim U(0,1)$ independent) has the moments $\frac{1}{n+1}$ of $U(0,1)$, so, being bounded, it is uniform, as Petrov conjectured.

Lemma 1. *$1 + \rho e^{i\theta} = R\,e^{ia\theta}$ for $0 \le \theta \le \pi$.*

Proof. As $b\theta = \theta - a\theta$: $\sin b\theta + \sin a\theta\, e^{i\theta} = \sin\theta\cos a\theta - \cos\theta\sin a\theta + \sin a\theta\cos\theta + i\sin a\theta\sin\theta = \sin\theta\, e^{ia\theta}$. Divide by $\sin b\theta$; at $\theta = 0$ both sides equal $1/b$. □

Lemma 2. *For $n \in \mathbb{N}_0$ and real κ, $\sum_{j=0}^n \binom{n}{j}\operatorname{sinc}(\kappa - j) = \binom{n}{\kappa}$.*

Proof. For $n = 0$ it reads $\operatorname{sinc}\kappa = 1/(\Gamma(1+\kappa)\Gamma(1-\kappa))$, the reflection formula. By Pascal's rule the left side $L_n(\kappa)$ satisfies $L_{n+1}(\kappa) = L_n(\kappa) + L_n(\kappa-1)$, and so does the right side, since $1/\Gamma(x) = x/\Gamma(x+1)$ gives $\binom{n}{\kappa} + \binom{n}{\kappa-1} = \frac{\Gamma(n+1)\,[(n+1-\kappa)+\kappa]}{\Gamma(\kappa+1)\Gamma(n+2-\kappa)} = \binom{n+1}{\kappa}$. Induct on n. □

Lemma 3. *For $n \in \mathbb{N}$, $\frac{1}{\pi}\int_0^\pi f^n\,\mathrm{d}\theta = \sum_{j=0}^n \binom{n}{j}\operatorname{sinc}(an - j)$.*

Proof. Let $\mu_j = j - an$, $\lambda = \rho'/\rho$ and $z^\mu := \rho^\mu e^{i\mu\theta}$, so $(z^\mu)' = \mu(\lambda + i)z^\mu$. By the binomial theorem and Lemma 1, $\sum_j \binom{n}{j} z^{\mu_j} = \rho^{-an}e^{-ian\theta}(1+\rho e^{i\theta})^n = \rho^{-an}R^n = f^n$. Hence

$$F = \sum_j \binom{n}{j}\frac{z^{\mu_j}}{\mu_j} \quad \left(\frac{z^0}{0} := \log\rho + i\theta\right) \qquad \text{has} \qquad F' = (\lambda + i)\,f^n.$$

As λ, f are real, $\Psi := \operatorname{Im} F = \sum_j \binom{n}{j}\rho^{\mu_j}\frac{\sin\mu_j\theta}{\mu_j}$ (with θ for $\mu_j = 0$) has $\Psi' = f^n$, so $\int_0^\pi f^n = \Psi(\pi) - \Psi(0) = \pi\sum_j \binom{n}{j}\operatorname{sinc}\mu_j$, since $\Psi(0) = 0$ and $\rho(\pi) = 1$; sinc is even. □

Proof of the Theorem. Let $h(x,t) = f(x)\,t^a(1-t)^b$ on $Q = [0,\pi]\times[0,1]$; it is continuous, so $0 \le h \le C$ for some $C \ge 1$. By Fubini and Euler's Beta integral, for real $r \ge 0$,

$$\frac{1}{\pi}\iint_Q h^r\,\mathrm{d}x\,\mathrm{d}t = \left(\frac{1}{\pi}\int_0^\pi f^r\,\mathrm{d}x\right)\frac{\Gamma(ar+1)\,\Gamma(br+1)}{\Gamma(r+2)}. \qquad (*)$$

For $r = n \in \mathbb{N}_0$, Lemmas 3 and 2 give $\frac{1}{\pi}\int_0^\pi f^n = \binom{n}{an}$ (trivially if $n = 0$), so $(*)$ equals $\frac{1}{n+1} = \int_0^1 v^n\,\mathrm{d}v$. Hence $\frac{1}{\pi}\iint_Q p(h) = \int_0^1 p$ for every polynomial p, and, approximating y^r uniformly on $[0,C] \supset [0,1]$ by polynomials (Weierstrass), $\frac{1}{\pi}\iint_Q h^r = \frac{1}{r+1}$. By $(*)$, $\frac{1}{\pi}\int_0^\pi f^r = \frac{\Gamma(r+2)}{(r+1)\Gamma(ar+1)\Gamma(br+1)} = \binom{r}{ar} = \binom{r}{s}$. □

Sources. Conjectured by T. Amdeberhan, MathOverflow Q253703 (2016). There F. Petrov gives the loop (integer n, k) and proposes the uniform reformulation; A. Selby's comment on Petrov's Nov. 19 answer extends the contour argument to complex n, k, outlining a proof for real r, s, and his answer gives an antiderivative in $y = \theta/n$ for even n (odd n similar) and real k, equal to $\Psi(ny)/n$. For real r, s the identity is implicit in Kanter's representation of stable laws (Ann. Probab. 1975, Cor. 4.1) and explicit as Devroye's normalizing constant (ACM TOMACS 2009); T. Simon (EJP 2014, Prop. 2, Rem. 3(b)) states it in binomial form for integer s. These rest on the Chernin–Ibragimov–Zolotarev integral representation of stable densities. New here is only this particular elementary assembly, which we did not find in the literature.